Project Proposal: Reproduction and Extension of Masked Autoencoders for Vision Learning

LAI CHEOK LAM, LIU YIFENG, LUO JIACHENG, LYU SHUJIE
TONG JIAQI, YANG LEYAN, ZHAI SHIAN

1 Research Paper Information

1.1 Paper Selection

Our project is based on the following paper: He, K., Chen, X., Xie, S., Li, Y., Dollár, P., and Girshick, R. (2022). Masked Autoencoders Are Scalable Vision Learners. CVPR 2022 [1].

Under the traditional supervised learning paradigm, the performance of visual models heavily relies on large-scale manually annotated datasets. However, obtaining such annotations is costly and difficult to scale. Meanwhile, existing self-supervised methods, such as contrastive learning, can leverage unlabeled data but typically depend on complex data augmentation strategies, large numbers of negative samples, and substantial computational resources, which limit their scalability and practical efficiency. Therefore, a central challenge in visual representation learning is how to learn feature representations with strong generalization ability from unlabeled data while maintaining high computational efficiency.

To address this challenge, the paper proposes a self-supervised learning framework based on Masked Image Modeling (MIM), known as the Masked Autoencoder (MAE). The core idea is to randomly mask a high proportion of image patches (e.g., 75%) and feed only the remaining visible patches into the encoder, while training the model to reconstruct the masked regions. This process forces the model to learn deep structural and semantic representations of the image. MAE adopts an asymmetric encoder–decoder architecture: the encoder processes only the visible patches, significantly reducing computational complexity, while a lightweight decoder combines the encoded features with mask tokens to reconstruct the full image. This design not only substantially improves training efficiency but also enhances the model's ability to capture global semantic structures.

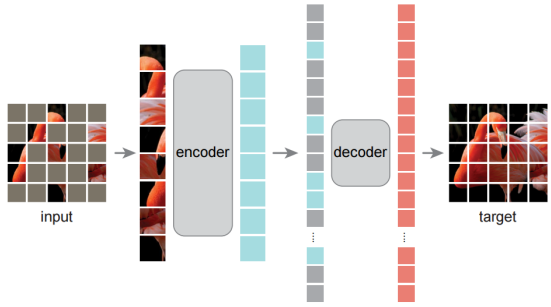

Fig. 1: Asymmetric encoder–decoder architecture of MAE. (Image from He et al. [1])

Experimental results demonstrate that MAE can efficiently learn general-purpose visual representations on large-scale datasets and significantly improve performance on downstream tasks such as image classification, object detection, and semantic segmentation after fine-tuning. These findings validate its effectiveness and superiority in scalable self-supervised visual learning.

1.2 Project Type and Objectives

Our project is categorized as Algorithm Reproduction and Application Extension. We will first reproduce the core architecture and training pipeline of MAE from scratch. Building upon this foundation, our extension will focus on two dimensions. From an algorithmic perspective, we aim to move beyond the conventional random geometric masking paradigm by exploring content-aware masking strategies, including Edge-Guided and Attention-Guided masking. From an application perspective, we will transfer the pretrained MAE encoder to the Oxford-IIIT Pet dataset for semantic segmentation fine-tuning. This is intended to systematically evaluate the model's ability to learn fine-grained local feature representations and its generalization performance in dense prediction tasks involving complex object boundaries. Ultimately, this project seeks to

deepen the understanding of MAE's underlying mechanisms and explore its optimization potential for specific downstream applications.

2 Background Reading Plan

To understand the fundamental principles and innovations of MAE, we plan to study the following related technical background:

Masked Autoencoders (MAE): The foundational baseline of this project. We will focus on studying its asymmetric encoder–decoder architecture and the experimental findings on random masking [1], which will serve as the reference baseline for the subsequent development of edge-guided and attention-guided masking strategies.

Vision Transformer (ViT): The architectural core of MAE. By studying its original paper [2], we aim to understand how the attention mechanism serializes image patches into sequences. This directly corresponds to the technical pathway of using attention weights to guide mask distribution in this project.

Semantic-aware Visual Pretraining: We will examine the evolution of masking strategies from random to semantic-aware approaches. By referring to BEiT [3] and similar non-uniform masking studies, we will analyze why masking difficult samples (such as edges or high-attention regions) forces the model to learn deeper structural representations.

Semantic Segmentation Architectures and Transfer Learning: To support the requirements of Direction B, we will study classical segmentation frameworks such as Mask R-CNN [4]. The goal is to understand how pretrained encoders can be integrated with downstream decoders and how to fine-tune models on fine-grained datasets such as Oxford-IIIT Pet.

3 Reproduction and Extension Plan

3.1 Reproduction Plan

To demonstrate substantial mastery of the model architecture, we will adopt an "implementation from scratch" engineering strategy. We will explicitly avoid directly using high-level model libraries (such as advanced APIs from timm or transformers). The implementation steps are as follows:

Bottom-level module construction: Fully based on the PyTorch framework, we will manually construct the ViT backbone modules (Block) containing Multi-head Self-Attention using basic linear layers and attention operations.

Core logic implementation: We will independently implement the random_masking function that controls the data flow, as well as the PatchEmbedding module.

Tensor dimension optimization: During the forward pass, we will use the einops library to handle all complex tensor reshaping operations. This improves code readability and ensures architectural modularity and decoupling (e.g., separating Encoder and Decoder).

Visualization probes: Hooks will be registered at key internal network nodes to record and export important intermediate information, such as attention maps, thereby enhancing the interpretability of the reproduced model.

3.2 Extension Plan

After successfully constructing and validating the baseline model, we will conduct extended research along two dimensions—masking mechanisms and downstream applications—within the available computational budget:

Direction A: Content-aware masking strategy exploration

The original paper evaluated three masking strategies: Random, Block, and Grid masking. These methods are fundamentally content-agnostic geometric occlusion strategies. To further explore the reconstruction limits and higher-level semantic reasoning capabilities of the model, we will move beyond traditional spatial occlusion paradigms and investigate two masking mechanisms based on the concept of hard example mining: Edge-Guided Masking and Attention-Guided Masking.

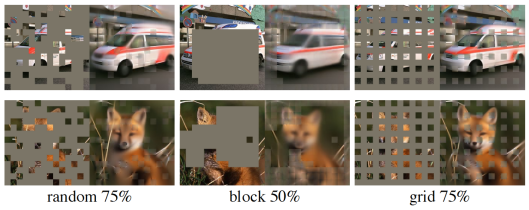

Fig. 2: Visualization comparison of different masking strategies. (Image adapted from He et al. [1])

Direction B: Downstream semantic segmentation task

To evaluate the efficacy of the pre-trained representations, we transfer the MAE encoder to the Oxford-IIIT Pet dataset for semantic segmentation fine-tuning. This dataset, characterized by intricate animal contours and

fine-grained hair textures, provides a rigorous test for the pixel-level local features learned by MAE.

4 Evaluation Plan

Due to computational constraints, we do not expect the model to reach the absolute accuracy reported in the original paper, which was trained for 800 epochs on the full dataset. Therefore, the evaluation framework of our project will focus on baseline comparison and relative performance improvements.

4.1 Qualitative Evaluation

Reconstruction capability visualization: We will visually compare the "original input image", the "masked image", and the "model reconstructed image" on the validation set. The focus will be on whether the model can reasonably infer the basic contours and colors of the missing regions under different masking strategies.

Fig. 3: Preliminary qualitative pretraining results (ImageNet-Mini, 100 epochs) indicate that the model has developed basic capabilities in local structural inference and color reconstruction.

Segmentation result comparison: In the downstream task, we will output the predicted segmentation masks on the Oxford-IIIT Pet dataset and examine whether the model can clearly delineate complex object boundaries, such as animal fur and body contours.

4.2 Quantitative Evaluation

Pre-training: We will continuously monitor and record the Mean Squared Error (MSE Loss) between reconstructed pixels and ground-truth pixels within the masked regions. This serves as a quantitative indicator of the model's information compression and reconstruction capability.

Fine-tuning for Segmentation: We will evaluate the mean Intersection over Union (mIoU) on the Oxford-IIIT Pet dataset. A randomly initialized ViT-Base model (trained from scratch) will be introduced as a strict baseline. By comparing the mIoU performance between the randomly initialized model and the MAE-pretrained model under identical fine-tuning settings, we aim to quantitatively demonstrate that MAE pre-training provides significant feature transfer advantages for dense prediction tasks, even under limited computational resources.

5 Division of Work

Our project is a collaborative effort by seven team members. To ensure the smooth progress of the project, the preliminary task allocation is shown in Table 1.

Table 1: Provisional Division of Work

Task	**Members**
Literature Review and Research	All Members
Proposal Writing	LYU SHUJIE, TONG JIAQI, YANG LEYAN
Model Reproduction (Core Code and Baseline Training)	2 persons
Extension A (Content-aware Masking Strategies)	2 persons
Extension B (Downstream Semantic Segmentation Fine-tuning)	2 persons

In addition to the specific responsibilities listed above, all members will be fully involved in daily discussions, the mid-term presentation, and the Final Report throughout the project.

References

[1] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked Autoencoders Are Scalable Vision Learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, et al. An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[3] Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. BEiT: BERT Pre-training of Image Transformers. In *International Conference on Learning Representations (ICLR)*, 2022.

[4] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.